

Convex optimization

Exercise 1

Consider a transmitter at a location $\mathbf{x}_t = (-2, -4)$. It is required to find the optimal location to place a receiver $\mathbf{x}_r$ within a specified area of land such that the received signal's power is maximized. The received signal's power is inversely proportional to the distance between the transmitter and the receiver:

$$P(\mathbf{x}) \propto \frac{1}{d} \quad (1)$$

The allowable area to place the receiver is given by the following constraints in the euclidean plane:

$$c_1(\mathbf{x}) = -x_1^2 - (x_2 + 4)^2 + 16 \geq 0 \quad (2)$$

$$c_2(\mathbf{x}) = x_1 - x_2 - 6 \geq 0 \quad (3)$$

- (a) Formulate the optimization problem (Write-up the cost function with the constraints). Is the **optimization problem** convex?
- (b) Consider now an additional constraint defined as:

$$c_3(\mathbf{x}) = x_1^2 + (x_2 + 6)^2 \geq 2. \quad (4)$$

Is the **optimization problem** convex?

Exercise 2

In this exercise, we will use the toolbox CVX with Matlab to solve the optimization problems in Exercise 1.

- (a) Make sure you have installed CVX with your matlab as provided in Moodle. (Download link <http://cvxr.com/cvx/download/>).
- (b) Solve for the best location of the receiver from Exercise 1 with the two constraints $c_1(\mathbf{x})$ and $c_2(\mathbf{x})$ and make sure the solution matches your graphical solution.
- (c) Try solving it for the worst location of the receiver. Does it work? If not then why?
- (d) Try solving for the best location of the receiver with the third constraint $c_3(\mathbf{x})$. Does it work? If not then why?

Exercise 3

In this exercise, we will perform linear regression to fit a function to the provided time series data. The data describes the measured concentration of CO₂ in parts per million at Hawaii over the time span between 1974 to 2020. For more information, you can open the file **co2_weekly_mlo.txt** and read the header.

The function which will be fitted in this exercise is of this structure.

$$f(t) = c_0 + c_1 f_1(t) + c_2 f_2(t) + \dots + c_n f_n(t) \quad (5)$$

where $f_i(t)$ is a base function and c_i is a linear coefficient for $f(t)$. The base functions are known and the coefficients will be solved for using the least squares method.

We will fit the function to the first half of the data and then see what happens when we extrapolate it into the future. Further, we will take a look at what can happen if the function is over fitted.

- (a) Run the file **co2_concentration_hawaii_ex.m**, visualize and investigate the data.
- (b) Find the sampling period of the measurements.
- (a) By investigating the data, choose a set of basis functions $f_i(t)$ for $i \in 1, \dots, n$ which could work as building blocks for the function.

Hint: The functions $\sin(2\pi ft)$ and $\cos(2\pi ft)$ are great at expressing periodic behavior at frequency f . The following figure shows the output from a fast fourier transform (FFT) of the data. An FFT is able to reveal important frequencies in the data.

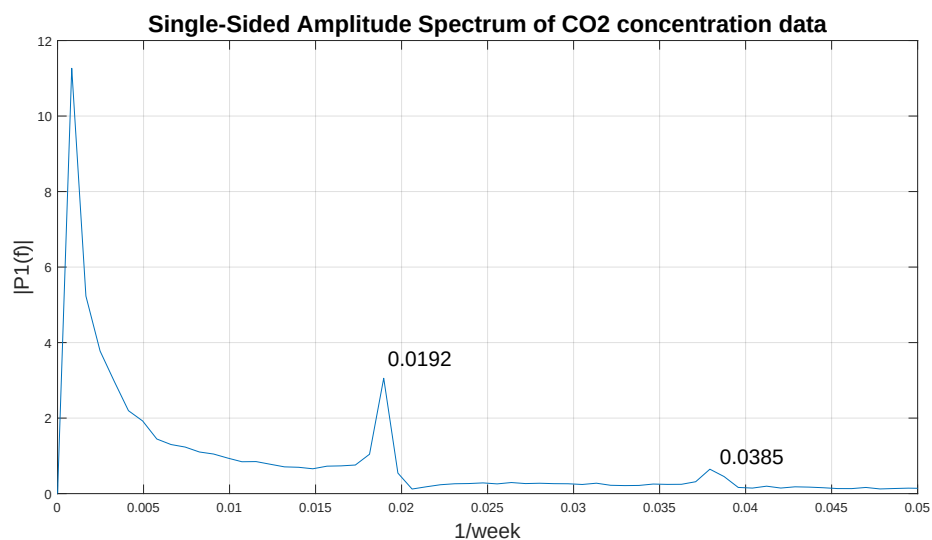


Figure 1: A FFT of the data provided in the Matlab table **data_fit**

- (d) Formulate the least squares problem for the data in the Matlab table **data_fit** and solve for the linear coefficients c_i .
- (e) Plot the result on top of the data. What do you see? Was the fit acceptable?
- (f) Extrapolate the function into the future and compare with the full set of measurements.
- (g) Create 100 different basis functions of your own choice and fit a new set of coefficients to the data in **data_fit**. What happens now when you extrapolate the function into the future?

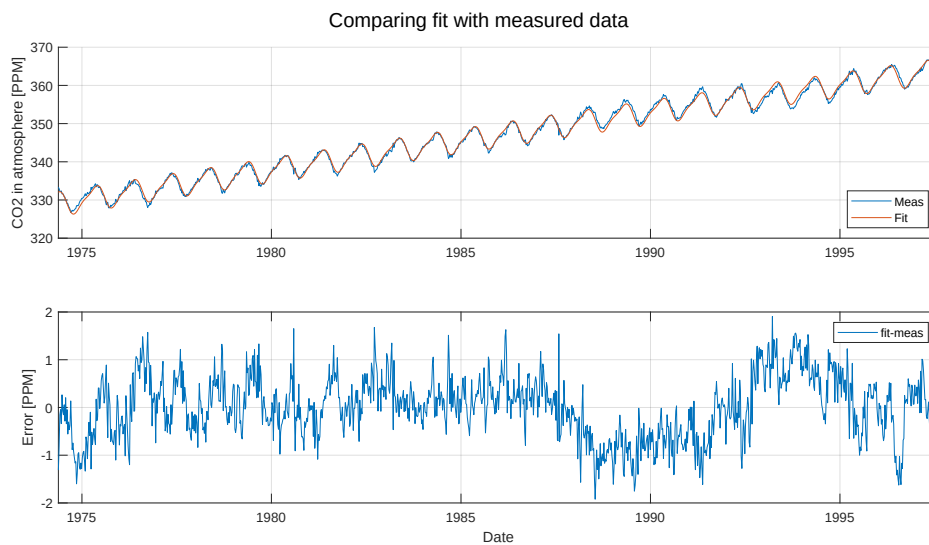


Figure 2: Shows our fitted function plotted on top of the data.